HEART DISEASE PREDICTOR

A Project Report
Submitted in partial fulfilment of the
Requirement for the award of the degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

Submitted By

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3008334

Under the esteemed guidance of
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S.S.T COLLEGE OF ARTS &COMMERCE

(Affiliated to University of Mumbai)

MAHARASHATRA

2024-2025

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2024-2025

CERTIFICATE

This is certify that the project entitled, "**HEART DISEASES PREDICTOR**", is Bonafide work of **Rachana Ramkumar Mahato** bearing seat no.(cs22053) submitted in partial fulfillment of the requirements for the award of degree of **DEPARTMENT OF SCIENCE (COMPUTER SCIENCE) from university of Mumbai.**

Internal Guide **coordinator**

External Examiner

Date: **College seal**

THE APPROVAL PROJECT PROPOSAL

(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

PNR No. _ Roll no.:**3008334**

1. Name of the student

   ______**Rachana Ramkumar Mahato**____________________

2. Title of the project

   ____**HEART DISEASE PREDICTOR**________________________

3. Name of the Guide

   _____**Asst. prof. Mr. Jeevan Rothe**____________________

4. Teaching experience of the Guide__________________________

5. Is this your first submission? Yes ☐ No ☐

Signature of the Student **Signature of the Guide**

Date: ………………… **Date: ………………**

Signature of the coordinator Date: …………………

ABSTRACT

Heart disease is a major global health concern and one of the leading causes of death worldwide. Early prediction and diagnosis play a crucial role in preventing severe outcomes and improving patient care. This project aims to develop a machine learning-based heart disease prediction system that can analyze key health indicators and accurately assess the risk of heart disease.

The model is trained using the UCI Heart Disease dataset, which includes features such as age, sex, chest pain type, blood pressure, cholesterol levels, and more. Various supervised learning algorithms including Logistic Regression, Random Forest, and Support Vector Machine are implemented and compared for performance.

The model is evaluated using accuracy, precision, recall, and ROC-AUC score. A simple and interactive user interface is also provided to allow real-time prediction based on user input.

This project demonstrates the potential of artificial intelligence in supporting medical professionals with early and efficient diagnosis of heart disease.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me throughout the completion of this project titled **"Heart Disease Prediction Using Machine Learning."**

First and foremost, I would like to thank my project guide, **[Asst. prof. Mr. Jeevan Rothe,** for their constant guidance, encouragement, and invaluable feedback throughout the course of this work. Their expertise and support have played a vital role in the successful execution of this project.

I am also grateful to the faculty members of the **[BS. COMPUTER SCIENCE]**, **[S.S.T COLLEGE OF ARTS & COMMERCE]**, for providing the necessary resources, knowledge, and motivation.

Special thanks to my classmates, friends, and family who supported me with their encouragement and constructive suggestions during every stage of the project.

Lastly, I would like to acknowledge the developers and contributors of the datasets and open-source tools that made this project possible.

This project has been a valuable learning experience, and I am thankful to everyone who contributed to its success.

DECLARATION

I hereby declare that the project entitled **“Heart Disease Prediction Using Machine Learning”** submitted in partial fulfillment of the requirements for the degree of **[Bachelor of Science In Computer Science]** is my original work and has not been submitted previously by me or any other individual for any other degree or diploma.

This project is the result of my independent efforts and has been carried out under the guidance of **[Mr. Jeevan Rothe]**, at **[S.S.T COLLEGE OF ARTS & COMMERCE]**.

All the information, data, and software used in this project have been properly acknowledged and referenced as standard academic ethics.

Place: Ulhasnagar

Date:

Signature of the Student
Rachana Ramkumar Mahato

TABLE OF CONTENTS

CHAPTER – 1

INTRODUCTION

Heart disease, also known as cardiovascular disease, is one of the leading causes of death globally. According to the World Health Organisation (WHO), millions of lives are lost each year due to heart-related conditions, many of which are preventable through early detection and timely treatment. With the growing amount of health-related data and the advancement of technology, the application of machine learning in healthcare has emerged as a promising approach for disease prediction and diagnosis.

This project focuses on developing a machine learning-based system that predicts the likelihood of heart disease in an individual based on various clinical parameters. These include age, gender, blood pressure, cholesterol levels, electrocardiogram results, and other relevant factors. The primary goal is to provide an accurate and efficient tool that can assist healthcare professionals in identifying high-risk patients at an early stage.

By leveraging historical medical data, the system trains predictive models using supervised learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine (SVM). These models are then evaluated using performance metrics like accuracy, precision, recall, and ROC-AUC to determine their effectiveness.

The implementation of this system demonstrates the potential of artificial intelligence in enhancing medical diagnosis and decision-making. It not only helps in reducing the burden on healthcare providers but also empowers individuals to take proactive steps toward managing their heart health.

1.1BACKGROUND

Heart disease is a major health concern that affects millions of people worldwide and contributes significantly to global mortality rates. In the medical community, cardiovascular diseases (CVDs) are considered among the most difficult to detect in early stages without advanced testing and expertise. Due to their complex nature and the involvement of various risk factors, both hereditary and lifestyle-based—timely diagnosis becomes crucial.

Traditionally, diagnosis of heart disease is based on a combination of clinical symptoms, laboratory tests, and diagnostic procedures such as Electrocardiograms (ECG), Echocardiography, Angiography, and Treadmill Tests. While these methods are accurate, they are often expensive, time-consuming, and require the involvement of experienced medical professionals. Additionally, manual interpretation of these test results can be prone to human error or bias.

In recent years, with the growing availability of healthcare data and advancements in computing power, machine learning has emerged as a promising tool for disease prediction. Machine learning algorithms can analyze large datasets, identify hidden patterns, and make predictions with high accuracy. In the context of heart disease, these models can be trained on patient data to recognize patterns associated with cardiovascular conditions and provide early warning signs.

Previous research has shown that machine learning models can outperform traditional statistical methods in predictive healthcare. Algorithms such as Logistic Regression, Decision Trees, Support Vector Machines, and Random Forests have been successfully used in disease classification tasks. Many of these models are capable of learning non-linear relationships among multiple variables, which is particularly useful in the medical domain where multiple interdependent risk factors are involved.

Publicly available datasets such as the Cleveland Heart Disease dataset from the UCI Machine Learning Repository have made it easier for researchers to build and test heart disease prediction models. These datasets typically include features such as age, sex, chest pain type, blood pressure, cholesterol level, and other diagnostic metrics.

This project builds on this foundation by implementing and comparing various machine learning algorithms to predict the likelihood of heart disease in individuals. The aim is to develop a reliable, user-friendly, and efficient system that can assist both patients and healthcare providers in making informed decisions.

1.2 Research Objectives

The primary objective of this project is to develop an intelligent system that can accurately predict the presence of heart disease in patients using machine learning techniques. The project aims to contribute to the field of healthcare by offering a data-driven solution that supports early diagnosis and improves clinical decision-making.

The specific objectives of the project are:

1. **To analyze and preprocess healthcare datasets** related to heart disease, ensuring data quality through techniques such as normalization, feature selection, and handling of missing values.

2. **To implement and compare multiple machine learning algorithms** (e.g., Logistic Regression, Random Forest, Support Vector Machine, etc.) for classification and prediction of heart disease.

3. **To evaluate the performance of the models** using key metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, in order to select the most effective algorithm.

4. **To develop a user-friendly interface** that allows users to input their health data and receive instant predictions regarding their risk of heart disease.

5. **To demonstrate the feasibility of using machine learning in healthcare** as a cost-effective and efficient tool for assisting medical professionals in early detection and risk assessment.

6. **To raise awareness about preventable cardiovascular risk factors** through visualization and feedback mechanisms that help users understand their health status.

1.3 PURPOSE, SCOPE, APPLICABILITY

1.3.1 PURPOSE:

The purpose of this project is to develop a machine learning-based system that can accurately predict the presence of heart disease in individuals based on various clinical and personal health attributes. Heart disease often goes undetected until it becomes life-threatening, and early diagnosis plays a vital role in reducing the risk of severe outcomes.

By leveraging historical healthcare data and machine learning algorithms, this system aims to:

- **Assist healthcare professionals** in identifying at-risk patients more efficiently.
- **Provide individuals with a preliminary risk assessment** based on their health inputs.
- **Reduce the dependency on expensive and time-consuming diagnostic procedures** by offering a fast, cost-effective, and accessible prediction tool.
- **Support preventive healthcare** by encouraging early intervention and lifestyle changes for individuals identified as high-risk.

Ultimately, the project seeks to demonstrate how artificial intelligence and data science can be applied in the healthcare domain to enhance disease detection, improve decision-making, and promote better health outcomes.

1.3.2 SCOPE:

The scope of this project focuses on the development and implementation of a machine learning-based system for predicting heart disease risk using clinical data. The project encompasses the following areas:

1. **Dataset Limitations**: The prediction model is trained using the publicly available **UCI Heart Disease dataset**, which includes specific features such as age, sex, blood pressure, cholesterol levels, and other health-related factors. The model's performance is therefore limited by the scope and quality of the available data.

2. **Feature Selection**: The system is designed to work with a set of predefined features (e.g., age, sex, cholesterol, blood pressure, etc.), meaning the prediction is based solely on the input parameters contained within the dataset. It does not account for all potential risk factors for heart disease, such as lifestyle habits (smoking, exercise, etc.) or genetic predispositions.

3. **Machine Learning Models**: The scope of the project includes the application of supervised learning algorithms, including Logistic Regression, Random Forest, and Support Vector Machine (SVM), to classify heart disease risk. The system is trained to predict whether an individual is at risk (1) or not at risk (0) based on the given features.

4. **Model Evaluation**: The evaluation of models is carried out using commonly used performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. These metrics help determine the effectiveness of each model and guide the selection of the best-performing one.

5. **User Interface**: The system will also include a basic user interface (UI), allowing users to input their health data and receive an instant prediction regarding their heart disease risk. The scope of the UI will be limited to data input and output prediction. It will not provide detailed medical recommendations or diagnoses.

6. **Deployment**: The project is limited to the development and testing of the model. While the model can be deployed on a web platform using tools like Flask or

Streamlit for user interaction, it is not intended to be used in real-time clinical environments without further validation and regulatory approvals.

7. **Geographical and Demographic Constraints**: The model is based on the dataset, which might not represent the entire global population, and it is primarily tailored to the demographic characteristics of the data. Thus, predictions may not be fully accurate for populations outside of the dataset's scope.

8. **Future Work**: The project does not include the implementation of advanced deep learning techniques, real-time monitoring, or integration with electronic health records (EHR). However, these can be part of potential future developments.

1.3.3 APPLICABILITY:

The machine learning-based **Heart Disease Prediction System** has broad applicability in various domains, particularly in healthcare, public health, and preventive medicine. Below are the key areas where this system can be applied:

1. **Healthcare Facilities and Clinics**:
   The system can be used by doctors and healthcare professionals to assist in the initial screening of patients. By providing a quick risk assessment, it helps prioritize individuals for further testing and diagnosis, improving the overall efficiency of healthcare services. It can be particularly beneficial in settings with limited access to advanced diagnostic technologies or specialized cardiologists.
2. **Remote and Rural Healthcare**:
   In rural or remote areas where healthcare resources and specialists may be scarce, this tool can provide an accessible way for individuals to assess their heart disease risk. It can be deployed on a simple web or mobile platform, enabling users to input their health data and receive immediate feedback without needing to travel long distances for consultation.
3. **Personal Health Monitoring and Preventive Care**:
   Individuals can use the system to assess their risk factors for heart disease and take preventive actions. It can encourage healthier lifestyles and proactive measures, such as dietary changes, physical activity, and regular medical check-ups. By empowering people to monitor their own health, it can play a role in reducing the global burden of heart disease.
4. **Public Health Campaigns**:
   Public health organizations can use this tool as part of awareness campaigns aimed at educating the population about heart disease risks. It can be used to promote preventive health measures and encourage individuals to undergo health check-ups, especially for those who might not realize they are at risk.
5. **Medical Research**:
   Researchers can use the system to analyze patterns and trends in heart disease prediction. By integrating it with other medical datasets or using it in longitudinal

studies, it can contribute valuable insights into the effectiveness of various risk factors and prediction methods.

6. **Corporate Health and Wellness Programs**:
   Employers and organizations focusing on employee well-being can integrate the heart disease prediction tool into their health initiatives. It can help identify at-risk employees and encourage early intervention, reducing healthcare costs and improving employee productivity.
7. **Integration with Wearable Devices (Future Scope)**:
   In the future, the system could be integrated with wearable health devices (e.g., fitness trackers, smartwatches) to provide real-time, personalized health assessments. This integration could allow for continuous monitoring of heart health indicators like heart rate and blood pressure, offering real-time alerts and advice.
8. **Cost-Effective Early Detection Tool**:
   The system offers a cost-effective solution for early heart disease detection compared to traditional diagnostic tests. In settings where advanced diagnostic tools are expensive or unavailable, this predictive model can serve as an initial screening tool before more advanced testing is carried out.
9. **Educational Purposes**:
   Educational institutions and organizations can use this tool as a case study in machine learning, data science, and healthcare technology. It demonstrates the practical application of machine learning algorithms in a real-world healthcare context.

1.4ORGANISATION OF REPORT:

The report is organized into several chapters to provide a comprehensive understanding of the project, its objectives, methodology, implementation, and outcomes. Each section is designed to provide the necessary details, from background information to the final conclusions and future work.

1. **Chapter 1: Introduction**
   - This chapter provides an overview of the project, including its motivation, objectives, scope, and purpose. It sets the foundation for understanding the importance of heart disease prediction and the role of machine learning in healthcare.

2. **Chapter 2: Literature Review**
   - In this chapter, the relevant research and existing works in the field of heart disease prediction using machine learning are reviewed. It discusses various methodologies, algorithms, and techniques previously applied to similar problems, highlighting their advantages and limitations.

3. **Chapter 3: Problem Statement**
   - This chapter clearly defines the problem being addressed in the project. It outlines the challenges in heart disease detection, the limitations of traditional diagnostic methods, and the need for an automated prediction system.

4. **Chapter 4: Methodology**
   - This chapter explains the approach taken to solve the problem, detailing the steps followed, from data collection to model evaluation. It includes data preprocessing, feature selection, machine learning algorithms implemented, and performance evaluation metrics.

5. **Chapter 5: System Design and Implementation**
   - In this chapter, the system architecture and design are presented. It describes the overall structure of the prediction system, including the user interface and backend components. It also explains the implementation of the machine learning models, training process, and integration with the user interface.

6. **Chapter 6: Results and Evaluation**
   - This chapter discusses the performance of the various machine learning models used in the project. It presents the results of the models, including accuracy, precision, recall, F1-score, and ROC-AUC.

7. **Chapter 7: Discussion**
    - In this section, the results are analyzed and discussed. It highlights the strengths and limitations of the developed system, provides insights into the implications of the findings, and compares the results with previous research.

8. **Chapter 8: Conclusion**
    - The final chapter provides a summary of the entire project, including key findings, conclusions, and recommendations. It also discusses the potential future improvements and enhancements to the system.

9. **References**
    - This section includes all the sources referenced throughout the report, including research papers, books, websites, and datasets.

10. **Appendices**
    - This section includes any additional information that supports the report, such as code snippets, additional charts, figures, or data tables that are relevant but not crucial to the main content.

CHAPTER – 2

SURVEY OF TECHNOLOGIES

The development of the **Heart Disease Prediction Using Machine Learning** system involves several technologies, including machine learning algorithms, programming languages, data preprocessing techniques, and web development tools. Below is a survey of the key technologies utilized in this project:

2.1. Machine Learning Algorithms

Machine learning algorithms are at the core of this project, enabling the system to make predictions based on data. The following machine learning techniques are used for heart disease prediction:

- **Logistic Regression**:
  Logistic Regression is a statistical method used for binary classification. It predicts the probability of an outcome (in this case, whether or not a person has heart disease) based on input features. It is a simple, interpretable, and effective algorithm for this type of classification problem.
- **Random Forest**:
  Random Forest is an ensemble learning method that builds multiple decision trees and merges them to improve the accuracy and robustness of predictions. It is particularly useful in handling complex datasets with multiple features and can capture non-linear relationships between variables.
- **Support Vector Machine (SVM)**:
  SVM is a supervised learning algorithm that finds the optimal hyperplane that best separates data into classes. It works well with high-dimensional datasets and is particularly effective in scenarios with clear margin of separation.
- **K-Nearest Neighbors (KNN)**:
  KNN is a non-parametric, lazy learning algorithm that classifies data based on the majority vote of the nearest neighbors. It is simple and effective, although computationally expensive when dealing with large datasets.

- **Naive Bayes**:
  Naive Bayes is a probabilistic classifier based on Bayes' Theorem, which assumes independence among predictors. It is widely used for classification tasks, especially when the dataset is large and the features are independent.

2.2. Programming Languages

The following programming languages were used in the development of the system:

- **Python**:
  Python is the primary programming language for this project, widely used in machine learning and data science due to its simplicity and a rich ecosystem of libraries and frameworks. Libraries such as pandas, numpy, scikit-learn, and matplotlib are used for data manipulation, machine learning, and visualization.
- **HTML/CSS**:
  HTML and CSS are used to design and structure the user interface of the application. HTML provides the markup for the web pages, while CSS is used to style the layout and design to make it user-friendly.
- **JavaScript**:
  JavaScript is used for adding interactivity to the front end of the application, especially if the system is deployed as a web application. It enables dynamic updates and asynchronous interactions with the server.

2.3. Machine Learning Libraries and Frameworks

Various machine learning and data science libraries were used to build and evaluate the predictive models:

- **scikit-learn**:
  scikit-learn is a powerful and easy-to-use Python library that provides a wide range of machine learning algorithms, including classification, regression, and clustering algorithms. It is used in this project to train and evaluate the models.
- **pandas**:
  pandas is an open-source data analysis library that provides high-level data structures and methods for efficiently manipulating and analyzing large datasets. It is used for data cleaning, preprocessing, and feature selection.

- **numpy**:
  numpy is a core scientific computing library in Python that provides support for large, multi-dimensional arrays and matrices. It is used for performing numerical operations on datasets and handling data efficiently.

- **matplotlib and seaborn**:
  matplotlib is a plotting library used for creating static, animated, and interactive visualizations in Python. seaborn is built on top of matplotlib and provides a higher-level interface for drawing attractive and informative statistical graphics. Both libraries are used for visualizing data and model performance.

2.4. Data Preprocessing Tools

Data preprocessing is crucial in ensuring the quality and usability of the dataset for model training. The following techniques and tools were applied:

- **Handling Missing Data**:
  Missing values are handled by imputation methods such as replacing with the mean, median, or mode, or using more advanced techniques like KNN imputation.

- **Normalization/Standardization**:
  Features with different units or scales are normalized or standardized to ensure that no single feature dominates the model training. Standardization methods like Z-score normalization or Min-Max scaling are used.

- **Feature Selection**:
  Feature selection is performed to choose the most relevant features for the prediction model, reducing dimensionality and improving performance. Techniques such as correlation matrices and recursive feature elimination (RFE) are used.

2.5. Web Development Tools

If the system is deployed as a web application, the following technologies are used for the user interface:

- **Flask**:
  Flask is a lightweight web framework for Python used to build web applications. It is used in this project to handle HTTP requests, interact with machine learning models, and serve the prediction results to users.

- **Streamlit** (optional):
  Streamlit is a framework used to create interactive web applications quickly. It is an excellent tool for creating user interfaces for machine learning models with minimal code.

2.6. Cloud Technologies (Optional)

For scalability and real-time prediction in production environments, cloud technologies may be used:

- **AWS, Google Cloud, or Azure**:
  Cloud platforms can host web applications, machine learning models, and datasets. They provide scalable resources for running models and deploying applications securely and efficiently.

CHAPTER – 3

REQUIREMENTS AND ANALYSIS

This section outlines the functional and non-functional requirements of the system, followed by a detailed analysis of how the system will work, including its inputs, outputs, and overall workflow.

3.1. Functional Requirements

Functional requirements define the core functions and behavior of the system. For this heart disease prediction system, they include:

- **Data Input**:
  The system must allow users to enter personal health parameters such as age, sex, blood pressure, cholesterol level, fasting blood sugar, maximum heart rate, etc.
- **Model Prediction**:
  The system should process the input data and predict whether the user is at risk of heart disease using a trained machine learning model.
- **Result Display**:
  The system should display the prediction result in a clear and user-friendly manner (e.g., "At Risk" or "Not At Risk").
- **Data Validation**:
  The system must validate user input to ensure all fields are correctly filled with valid values.
- **User Interface (UI)**:
  A simple and intuitive interface must be provided to interact with the model and view results.
- **Model Comparison (Optional)**:
  The system can include a comparison feature to show the performance of different models used in the backend.

3.2. Non-Functional Requirements

Non-functional requirements define the quality attributes of the system:

- **Accuracy**:
  The system should provide high accuracy in predictions, preferably above 85%, depending on the model performance.
- **Performance**:
  The prediction process must be quick and responsive, providing results within a few seconds.
- **Scalability**:
  The system should be scalable to support additional data or model upgrades in the future.
- **Usability**:
  The interface must be user-friendly for both technical and non-technical users.
- **Maintainability**:
  The system should be easy to update or modify in case new models or features are added.
- **Security**:
  The system must securely handle user inputs, especially if deployed online, to prevent misuse or data leakage.

3.3. System Inputs

The system requires the following user health parameters (based on UCI Heart Disease dataset):

- Age
- Sex
- Chest Pain Type (cp)
- Resting Blood Pressure (trestbps)

- Serum Cholesterol (chol)
- Fasting Blood Sugar (fbs)
- Resting ECG Results (restecg)
- Maximum Heart Rate Achieved (thalach)
- Exercise-Induced Angina (exang)
- ST Depression (oldpeak)
- Slope of Peak Exercise ST Segment (slope)
- Number of Major Vessels Colored (ca)
- Thalassemia (thal)

3.4. System Outputs

- A classification result:
  - **0** = Not at risk of heart disease
  - **1** = At risk of heart disease
- Optional: Display probability/confidence score for the prediction (e.g., "You have a 78% chance of being at risk").

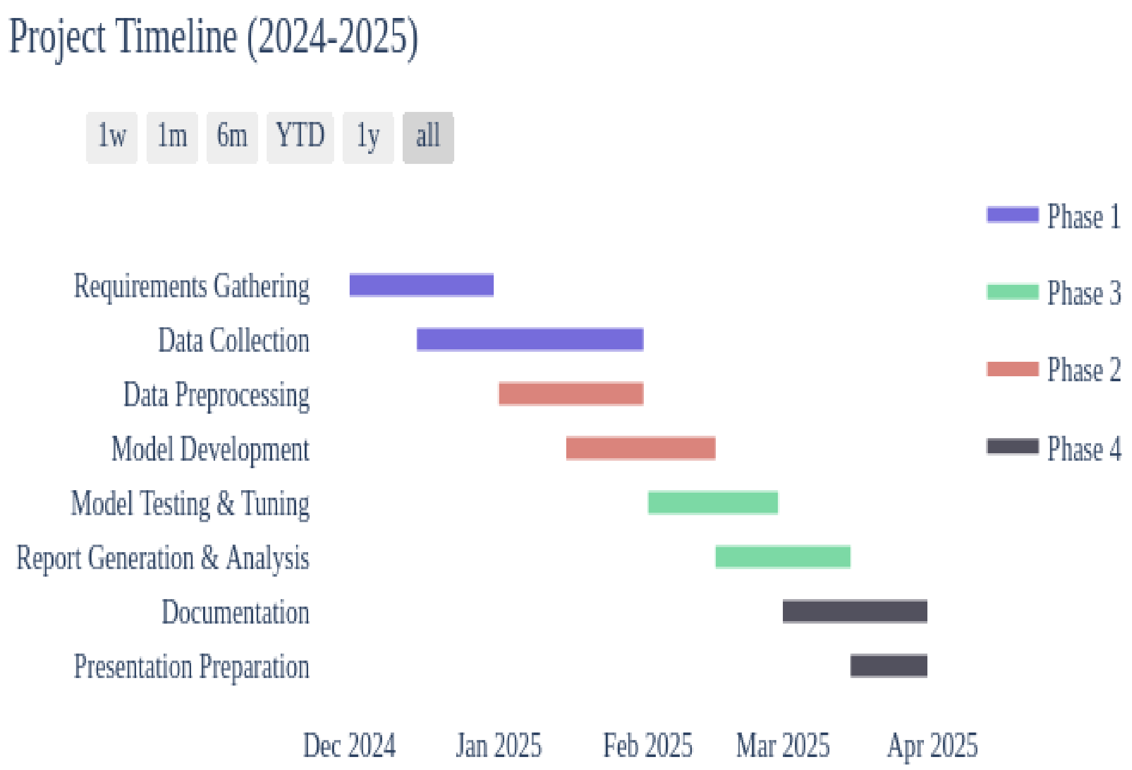
3.5. System Analysis

The system follows a typical machine learning pipeline:

1. **Data Collection**:
   The UCI Heart Disease dataset is used, containing labeled data of patients with and without heart disease.
2. **Data Preprocessing**:
   Missing values are handled, categorical values are encoded, and numerical features are scaled or normalized.

3. **Model Training**:
   Several algorithms (Logistic Regression, Random Forest, SVM, etc.) are trained using the training portion of the dataset.
4. **Model Evaluation**:
   Accuracy, precision, recall, F1-score, and ROC-AUC metrics are used to evaluate each model's performance.
5. **Prediction Module**:
   The best-performing model is selected and used for real-time prediction when a user inputs their data.
6. **Interface Integration**:
   The trained model is integrated with a web-based or desktop interface to enable user interaction.

CHART TITLE


Project Timeline (2024-2025)
1w
1m
6m
YTD
1y
all
Phase 1
Phase 3
Phase 2
Phase 4
Requirements Gathering
Data Collection
Data Preprocessing
Model Development
Model Testing & Tuning
Report Generation & Analysis
Documentation
Presentation Preparation
Dec 2024
Jan 2025
Feb 2025
Mar 2025
Apr 2025

3.6 Software and Hardware Requirements

This section outlines the necessary software and hardware components required for the development, testing, and execution of the heart disease prediction system.

3.6.1. Software Requirements

Software Component	Description
Operating System	Windows 10 or higher / Linux / macOS
Programming Language	Python 3.8 or above
IDE/Code Editor	Jupyter Notebook / VS Code / PyCharm
Libraries & Frameworks	pandas, numpy, scikit-learn, matplotlib, seaborn, joblib or pickle (for model saving), Flask or Streamlit (for deployment)
Browser	Google Chrome / Mozilla Firefox (for web-based UI)
Database (Optional)	SQLite or MySQL (if user data needs to be stored)
Version Control	Git (for source code management)

3.6.2. Hardware Requirements

Hardware Component	Minimum Requirement	Recommended
Processor	Intel Core i3 or equivalent	Intel Core i5/i7 or AMD Ryzen 5+
RAM	4 GB	8 GB or higher

Hardware Component	Minimum Requirement	Recommended
Hard Disk	500 MB (for project files and dependencies)	1 GB+
Display	1024x768 resolution	1366x768 or higher
Internet	Required for package installation and model deployment (if using cloud)	High-speed recommended

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Optional Cloud Requirements (for deployment)

Service	Description
Heroku / Render / Railway	Free or low-cost platforms to deploy small Flask/Streamlit apps
Google Colab	Cloud-based environment for running and testing models
AWS/GCP/Azure	Advanced deployment and scalability (optional for enterprise-level apps)

3.7 Preliminary Product Description

Product Overview

The system takes input from users in the form of personal and clinical attributes such as age, gender, blood pressure, cholesterol levels, chest pain type, and other key factors. Using a trained machine learning model (e.g., Random Forest, Logistic Regression, etc.), it predicts whether a person is at risk of heart disease.

The system aims to be:

- **Accurate** in predictions.
- **User-friendly** with a simple interface.

- **Quick** to deliver results.
- **Accessible** either as a standalone desktop app or a lightweight web application.

3.7.1 Key Features

- **Input Form**: A clean user interface to input patient health details.
- **Prediction Engine**: Uses a trained ML model to assess risk based on inputs.
- **Result Display**: Presents a clear outcome ("At Risk" / "Not At Risk") with optional confidence score.
- **Model Performance**: Behind the scenes, the model is evaluated and chosen based on performance metrics like accuracy, precision, and recall.
- **Optional Visualization**: Charts or graphs showing model performance or data distribution.
- **Scalability**: Can be expanded to include more features or be integrated with electronic health record systems.

3.7.2 User Roles

- **General Users / Patients**: Can input personal health data to assess risk level.
- **Doctors / Healthcare Professionals**: Can use the system as a screening tool to assist in diagnosis.
- **Researchers / Students**: Can use the system for academic or analytical purposes.

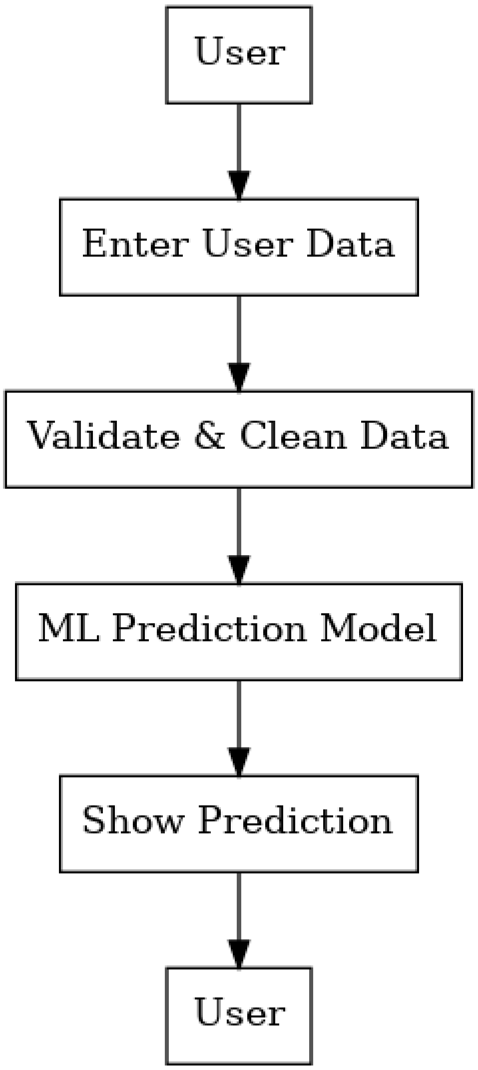
3.7.3Product Benefits

- **Early Detection**: Helps in identifying risk factors early, enabling timely treatment.
- **Cost-Effective**: Reduces the need for expensive tests for preliminary screening.
- **Educational Tool**: Useful for teaching machine learning applications in healthcare.

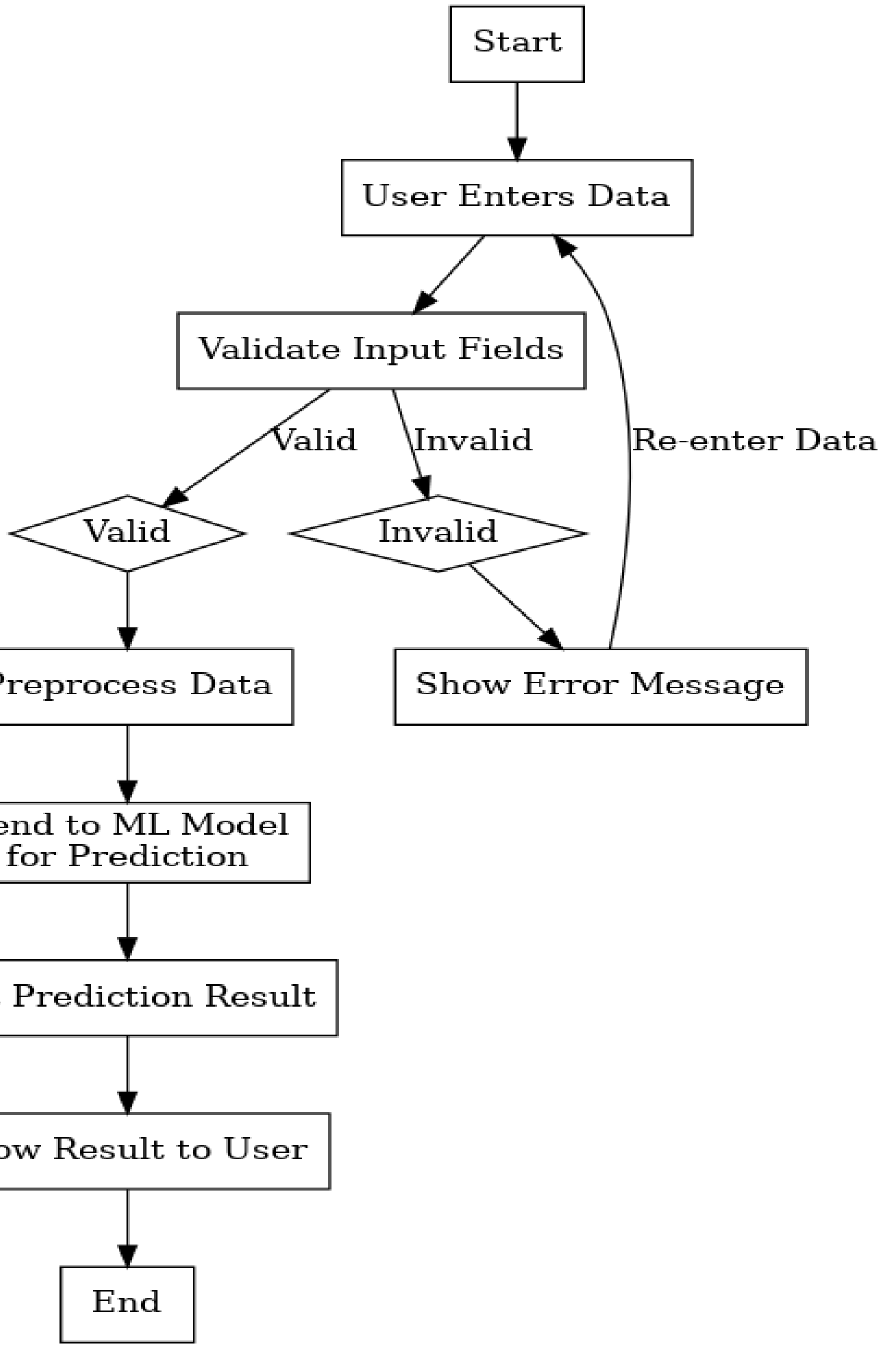
3.8 Conceptual Models

Conceptual models provide a high-level view of how the system works, outlining the flow of data, system components, user interactions, and overall architecture. These models help in understanding the logic and design of the heart disease prediction system.

Data Flow Diagram (Level 0).


User
Enter User Data
Validate & Clean Data
ML Prediction Model
Show Prediction
User

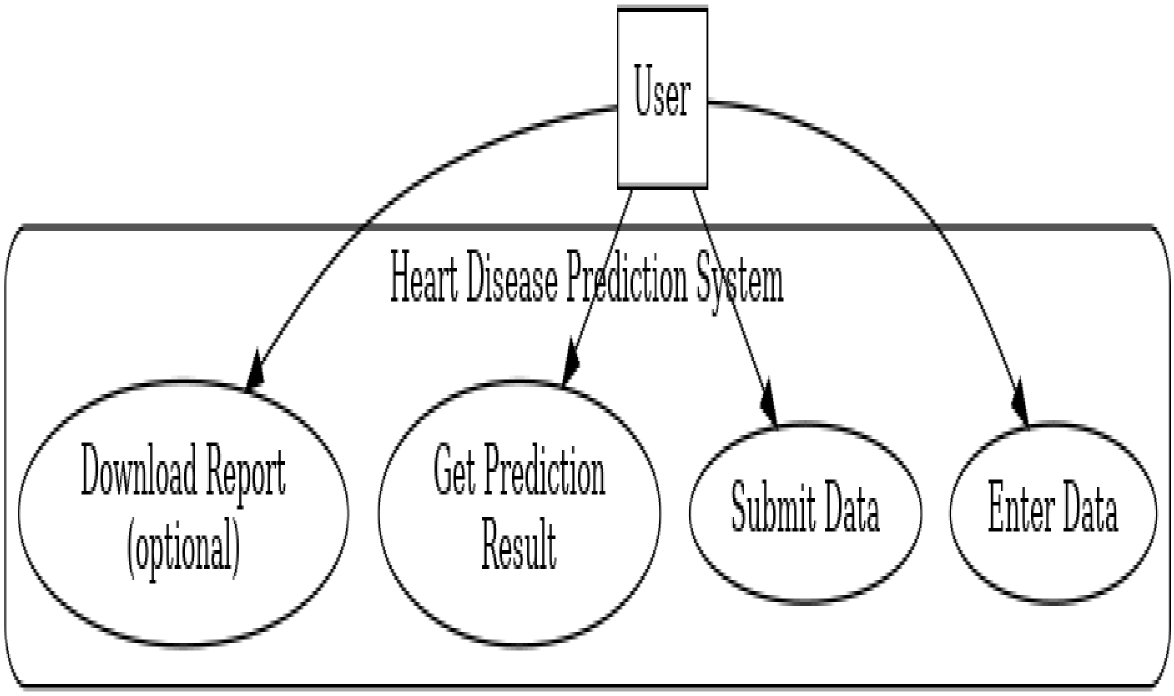
DFD(level1)


Start
User Enters Data
Validate Input Fields
Valid
Invalid
Re-enter Data
Valid
Invalid
Preprocess Data
Show Error Message
Send to ML Model
for Prediction
Get Prediction Result
Show Result to User
End

FLOW CHART

Use Case Diagram and Description

Use Case diagrams visually describe how users interact with the system and what functionalities are available to them. The primary actor in this system is the **User** (which could be a patient or healthcare provider).



Use Case Diagram

Chapter -4

System Design

System Design

The **System Design** phase is crucial in defining the architecture, components, modules, interfaces, and data flow of the Heart Disease Prediction System. It transforms the conceptual models into a structured framework that can be implemented during development.

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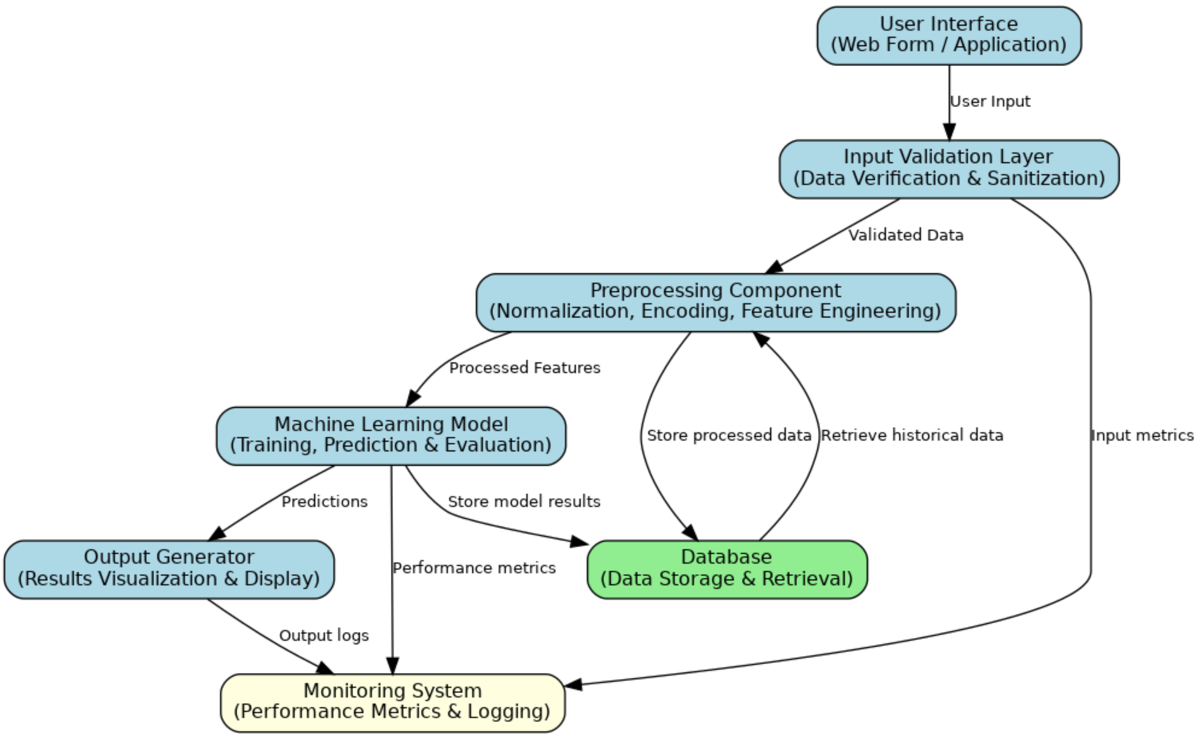
1. System Architecture

The architecture of the heart disease prediction system follows a modular structure with a clear separation between the frontend, backend, and machine learning components.



2. Major System Modules

a) User Input Module

- Accepts user data (age, gender, blood pressure, cholesterol, etc.)
- Provides dropdowns or text fields for easy input
- Performs basic client-side validation

b) Preprocessing Module

- Handles missing values
- Applies feature scaling (standardization or normalization)
- Encodes categorical variables (e.g., chest pain type, thalassemia)

c) Prediction Module (ML Engine)

- Loads trained machine learning model (e.g., RandomForest.pkl)
- Accepts preprocessed input
- Outputs risk classification (0 = Not at Risk, 1 = At Risk)
- Optionally displays probability/confidence score

d) Result Display Module

- Displays prediction in a user-friendly format
- Optional: Includes visual aids (e.g., color coding, probability bars)

e) Optional: Database Module

- Stores user input and prediction result
- Helps in audit/logging or longitudinal tracking (if needed)

3. Data Flow Design

1. **User** fills in health details via a web form.
2. Input data is sent to the **backend (Flask/Streamlit)**.
3. **Backend** validates and preprocesses the data.
4. The **ML model** takes the data and returns the prediction.
5. The prediction is **displayed to the user** in real time.

---

4. Model Deployment Design

- **Local Deployment**: Using Streamlit or Flask for local usage.
- **Web Deployment**: Hosted on Heroku, Render, or similar cloud platforms.
- **Model Loading**: ML model is saved as .pkl file and loaded at runtime for predictions.

---

5. Frontend Design (UI/UX)

- Simple, clean layout with form fields.
- Clear button for "Predict".
- Display section for showing results.
- Optional: Informational tooltips or error messages for incorrect input.

Chapter -5
Implementation &Testing

- **Prediction.py:**

```
# importing required libraries

import numpy as np

import pandas as pd

import pickle

from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

from sklearn.ensemble import RandomForestClassifier

from sklearn.svm import SVC

from sklearn.linear_model import LogisticRegression

from sklearn.neighbors import KNeighborsClassifier

from sklearn.tree import DecisionTreeClassifier


# loading and reading the dataset


heart = pd.read_csv("C:\\Users\\teana\\Downloads\\Heart-Disease-Prediction-Deployment-master\\Heart-Disease-Prediction-Deployment-master\\heart_cleveland_upload.csv")


# creating a copy of dataset so that will not affect our original dataset.
```

```
heart_df = heart.copy()

# Renaming some of the columns
heart_df = heart_df.rename(columns={'condition':'target'})
print(heart_df.head())

# model building

#fixing our data in x and y. Here y contains target data and X contains rest all the features.
x= heart_df.drop(columns= 'target')
y= heart_df.target

# splitting our dataset into training and testing for this we will use train_test_split library.
x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=42)

#feature scaling
scaler= StandardScaler()
x_train_scaler= scaler.fit_transform(x_train)
x_test_scaler= scaler.fit_transform(x_test)

# creating K-Nearest-Neighbor classifier
model=RandomForestClassifier(n_estimators=20)
model.fit(x_train_scaler, y_train)
y_pred= model.predict(x_test_scaler)
```

```
p = model.score(x_test_scaler,y_test)
print(p)

print('Classification Report\n', classification_report(y_test, y_pred))
print('Accuracy: {}%\n'.format(round((accuracy_score(y_test, y_pred)*100),2)))

cm = confusion_matrix(y_test, y_pred)
print(cm)

# Creating a pickle file for the classifier
filename = 'heart-disease-prediction-knn-model.pkl'
pickle.dump(model, open(filename, 'wb'))
```

- **APP.PY:**

```
# Importing essential libraries
from flask import Flask, render_template, request
import pickle
import numpy as np


# Load the Random Forest CLassifier model
filename = 'heart-disease-prediction-knn-model.pkl'
model = pickle.load(open(filename, 'rb'))


app = Flask(__name__)


@app.route('/')
def home():
    return render_template('main.html')


@app.route('/predict', methods=['GET','POST'])
def predict():
    if request.method == 'POST':


        age = int(request.form['age'])
        sex = request.form.get('sex')
        cp = request.form.get('cp')
```

```
        trestbps = int(request.form['trestbps'])
        chol = int(request.form['chol'])
        fbs = request.form.get('fbs')
        restecg = int(request.form['restecg'])
        thalach = int(request.form['thalach'])
        exang = request.form.get('exang')
        oldpeak = float(request.form['oldpeak'])
        slope = request.form.get('slope')
        ca = int(request.form['ca'])
        thal = request.form.get('thal')

        data =
np.array([[age,sex,cp,trestbps,chol,fbs,restecg,thalach,exang,oldpeak,slope,ca,thal]])
        my_prediction = model.predict(data)

        return render_template('result.html', prediction=my_prediction)



if __name__ == '__main__':
    app.run(debug=True)
```

- **MAIN.HTML:**

```
<!DOCTYPE html>


<html lang="en" dir="ltr">


<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Heart Disease Predictor</title>
  <link rel="stylesheet" type="text/css" href="{{ url_for('static', filename='style.css') }}">
  <script src="https://kit.fontawesome.com/5f3f547070.js"
crossorigin="anonymous"></script>
  <link href="https://fonts.googleapis.com/css2?family=Pacifico&display=swap"
rel="stylesheet">
</head>


<body>


  <!-- Website Title -->


  <div class="container">
    <h2 class='container-heading'><span class="heading_font">Heart Disease
Predictor</span></h2>


  </div>
```

```
<!-- Text Area -->
<div class="ml-container">
  <form action="{{ url_for('predict')}}" method="POST">


      <label for="age">Age</label>
      <input type="text" id="age" name="age" placeholder="Your age.."><br>


      <label for="sex">Sex</label>
      <select id="sex" name="sex">
        <option selected>----select option----</option>
        <option value="1">Male</option>
        <option value="0">Female</option>
      </select><br>


      <label for="cp">Chest Pain Type</label>
      <select id="cp" name="cp">
        <option selected>----select option----</option>
        <option value="0">Typical Angina</option>
        <option value="1">Atypical Angina</option>
        <option value="2">Non-anginal Pain</option>
        <option value="3">Asymtomatic</option>
```

```
</select><br>

<label for="trestbps">Resting Blood Pressure</label>
<input type="text" id="trestbps" name="trestbps" placeholder="A number in range [94-
200] mmHg"><br>

<label for="chol">Serum Cholesterol</label>
<input type="text" id="chol" name="chol" placeholder="A number in range [126-564]
mg/dl"><br>

<label for="fbs">Fasting Blood Sugar</label>
<select id="fbs" name="fbs">
  <option selected>----select option----</option>
  <option value="1">Greater than 120 mg/dl</option>
  <option value="0">Less than 120 mg/dl</option>
</select><br>

<label for="restecg">Resting ECG Results</label>
<select id="restecg" name="restecg">
  <option selected>----select option----</option>
  <option value="0">Normal</option>
  <option value="1">Having ST-T wave abnormality</option>
```

```
  <option value="2">Probable or definite left ventricular
    hypertrophy</option>
</select><br>

<label for="thalach">Max Heart Rate </label>
<input type="text" id="thalach" name="thalach" placeholder="A number in range [71-
202] bpm"><br>

<label for="exang">Exercise-induced Angina</label>
<select id="exang" name="exang">
  <option selected>----select option----</option>
  <option value="1">Yes</option>
  <option value="0">No</option>
</select><br>

<label for="oldpeak">ST depression</label>
<input type="text" id="oldpeak" name="oldpeak" placeholder="ST depression, typically
in [0-6.2]"><br>

<label for="slope">slope of the peak exercise ST segment</label>
<select id="slope" name="slope">
  <option selected>----select option----</option>
  <option value="0">Upsloping</option>
```

```html
        <option value="1">Flat</option>
        <option value="2">Downsloping</option>
      </select><br>


      <label for="ca">Number of Major vessels</label>
      <input type="text" id="ca" name="ca" placeholder="Typically in [0-4]"><br>


      <label for="thal">Thalassemia</label>
      <select id="thal" name="thal">
        <option selected>----select option----</option>
        <option value="0">Normal</option>
        <option value="1">Fixed Defect</option>
        <option value="2">Reversible Defect</option>
      </select><br>



     <input type="submit" class="my-cta-button" value="Predict">
    </form>
  </div>
</body>
</html>
```

- **RESULTS.HTML :**

```
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Heart Disease Predictor</title>
    <link rel="shortcut icon" href="{{ url_for('static', filename='diabetes-favicon.ico') }}">
    <link rel="stylesheet" type="text/css" href="{{ url_for('static', filename='style.css') }}">
    <script src="https://kit.fontawesome.com/5f3f547070.js"
crossorigin="anonymous"></script>
    <link href="https://fonts.googleapis.com/css2?family=Pacifico&display=swap"
rel="stylesheet">
  </head>

  <body>

    <!-- Website Title -->
    <div class="container">
      <h2 class='container-heading'><span class="heading_font">Heart Disease
Predictor</span></h2>

    </div>

    <!-- Result -->
```

```
<div class="results">
    {% if prediction==1 %}
        <h1>Prediction: <span class='danger'>yes Heart Disease.</span></h1>

    {% elif prediction==0 %}
        <h1>Prediction: <span class='safe'>no Heart Disease.</span></h1>

    {% endif %}
</div>
</body>
</html>
```

- **STYLE.CSS :**

```
html{
    height: 100%;
    margin: 0;
}

body{
    font-family: Arial, Helvetica,sans-serif;
    text-align: center;
    margin: 0;
    padding: 0;
    width: 100%;
    height: 100%;
    display: flex;
    flex-direction: column;
}

/* Website Title */
.container{
    padding: 30px;
    position:relative;
    background: linear-gradient(45deg, #161616, #383436, #161616);
    background-size: 500% 500%;
    animation: change-gradient 10s ease-in-out infinite;
```

```
}
@keyframes change-gradient {
    0%{
        background-position: 0 50%;
    }
    50%{
        background-position: 100% 50%;
    }
    100%{
        background-position: 0 50%;
    }
}
.container-heading{
    margin: 0;
}
.heading_font{
    color: #ffffff;
    font-family: 'Pacifico', cursive;
    font-size: 35px;
    font-weight: normal;
}

.description p{
    color: #ffffff;
```

```
    font-style: italic;
    font-size: 14px;
    margin: -5px 0 0;
}
/* Text Area */
.ml-container{
    margin: 30px 0;
    flex: 1 0 auto;
}
.form {
    text-align: center;
    width: 250px;
    height: 25px;
    margin-bottom: 5px;

}
input[type=text], select {
    width:60%;
    padding: 12px 20px;
    margin: 8px 0;
    display: inline-block;
    border: 1px solid #ccc;
    border-radius: 4px;
    box-sizing: border-box;
```

```
}

label {
    display: inline block;
    width: 200px;
    font-weight: bold;
    text-align: center;
    float: left;

  }
/* Predict Button */
.my-cta-button{
    background: #f9f9f9;
    border: 2px solid #000000;
    border-radius: 1000px;
    box-shadow: 3px 3px #8c8c8c;
    margin-top: 10px;
    padding: 10px 36px;
    color: #000000;
    display: inline-block;
    font: italic bold 20px/1 "Calibri", sans-serif;
    text-align: center;
}
.my-cta-button:hover{
```

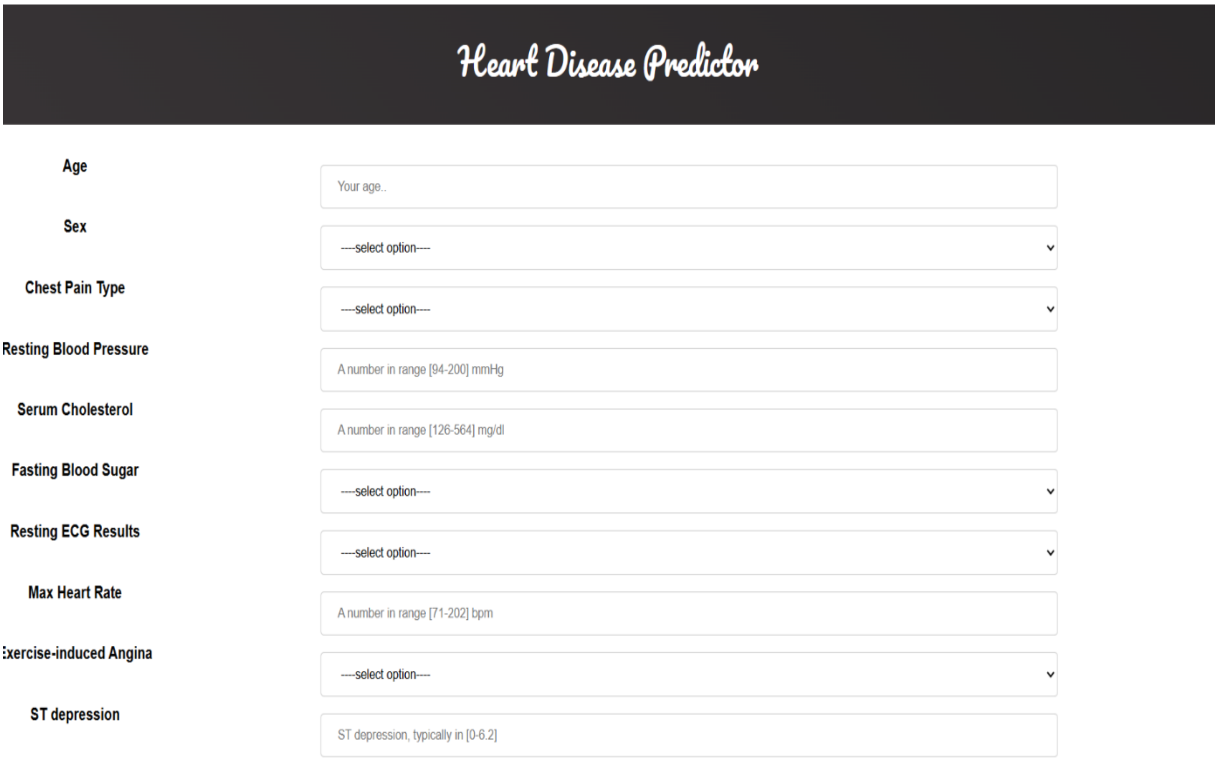
```
    color: #141414;
    border: 2px solid #46424b;
}
.my-cta-button:active{
    box-shadow: 0 0;
}
/* Contact */
.contact-icon{
    color: #ffffff;
    padding: 7px;
}
.contact-icon:hover{
    color: #8c8c8c;
}
/* Footer */
.footer{
    flex-shrink: 0;
    position:relative;
    padding: 20px;
    background: linear-gradient(45deg, #161616, #383436, #161616);
    background-size: 500% 500%;
    animation: change-gradient 10s ease-in-out infinite;
}
.footer-description{
```

```
    color: #ffffff;
    margin: 0;
    font-size: 12px;
}
/* Result */
.results{
    padding: 30px 0 0;
    flex: 1 0 auto;
}

.danger{
    color: #ff0000;
}

.safe{
    color: green;
}
```

- **TESTING:-**


Heart Disease Predictor
Age
Your age..
Sex
----select option----
Chest Pain Type
----select option----
Resting Blood Pressure
A number in range [94-200] mmHg
Serum Cholesterol
A number in range [126-564] mg/dl
Fasting Blood Sugar
----select option----
Resting ECG Results
----select option----
Max Heart Rate
A number in range [71-202] bpm
Exercise-induced Angina
----select option----
ST depression
ST depression, typically in [0-6.2]


Heart Disease Predictor
Prediction: yes Heart Disease.

CONCLUSION

The development of the Heart Disease Prediction System using Machine Learning provides a powerful, accessible, and efficient tool for the early detection of heart disease. By leveraging historical medical data and applying supervised learning algorithms, the system is capable of predicting the likelihood of heart disease in individuals based on key health parameters.

The system simplifies the diagnostic process by analyzing patterns in user input data such as age, blood pressure, cholesterol, chest pain type, and other attributes. It offers real-time risk assessment, which can aid healthcare professionals in making quicker, more informed decisions and promote preventive care in patients.

Throughout this project, machine learning techniques like Logistic Regression, Random Forest, and Support Vector Machines were explored and evaluated. After thorough testing, the best-performing model was selected based on accuracy and other performance metrics. The integration of this model into a user-friendly interface (web or desktop) makes the tool practical for both medical professionals and the general public.

In conclusion, this project demonstrates how data science and machine learning can significantly contribute to the healthcare sector by enabling early prediction, cost-effective diagnosis, and improved patient outcomes. With future enhancements—such as real-time data integration, cloud-based deployment, and support for larger datasets—this system has the potential to become an even more robust diagnostic assistant.

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